

Diabetes Mellitus and Pancreatitis

Learning Objectives

- ⇒ To have a brief understanding of diabetes mellitus and pancreatitis
- ⇒ To know about the clinical features of diabetes mellitus, pancreatitis
- ⇒ To know about the complications of diabetes mellitus and pancreatitis
- ⇒ To know about the physical examination to be performed in the patients of diabetes mellitus and pancreatitis
- ⇒ To know about ordering of laboratory investigations to be performed in above mentioned diseases and interpret the results
- ⇒ To know about pancreatitis induced diabetes mellitus (fibrocalculous pancreatic diabetes mellitus)

Diabetes Mellitus

Diabetic Complications

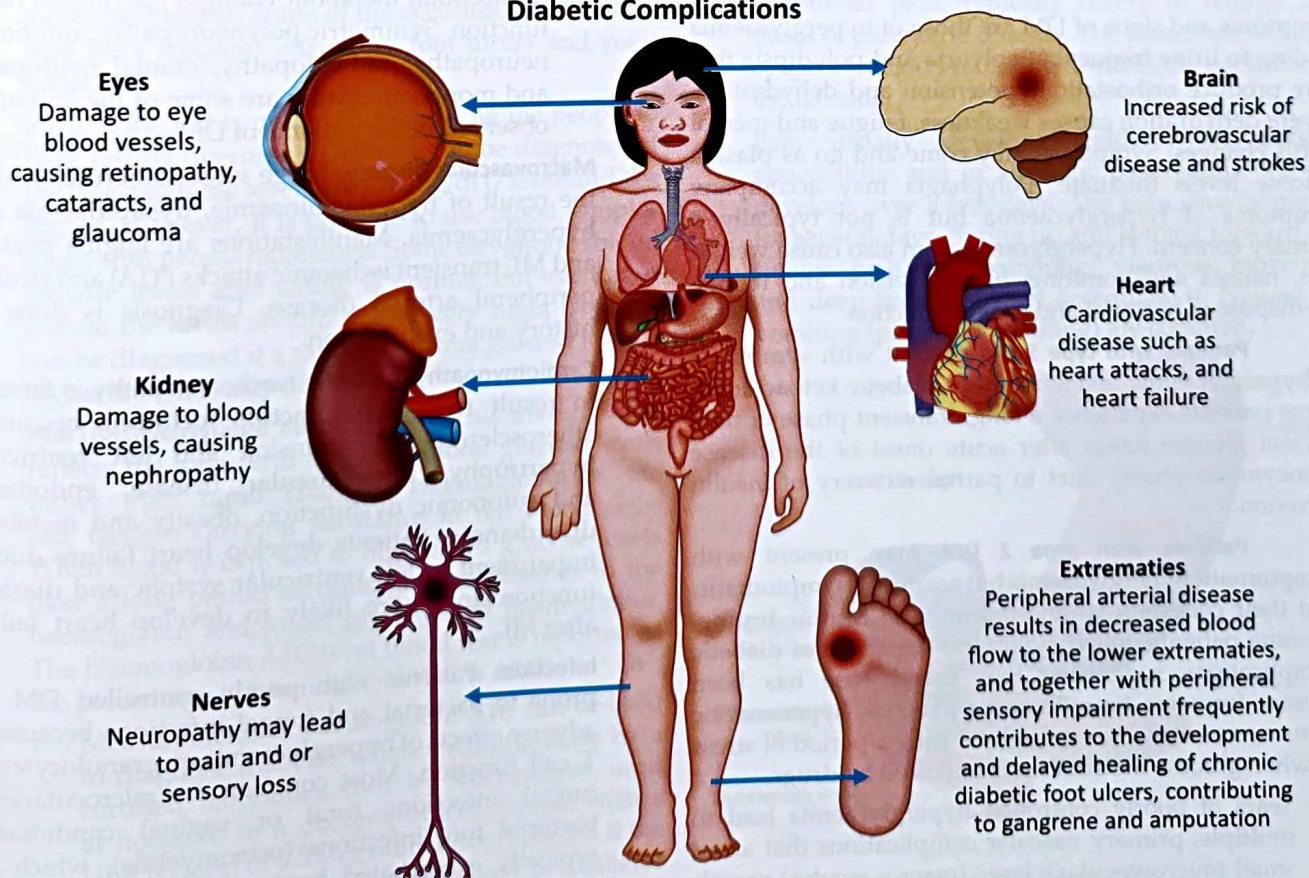


Figure 163: Complications of diabetes mellitus

Diabetes mellitus (DM) is impaired insulin secretion and variable degrees of peripheral insulin resistance leading to hyperglycaemia. Early symptoms are related to hyperglycaemia and include polydipsia (excessive thirst), polyphagia (excess eating) and polyuria (passage of excess amount of urine). Later complications include vascular diseases, peripheral neuropathy and predisposition to infection. Diagnosis is by measuring plasma glucose.

Types of Diabetes Mellitus

Two types of DM are observed Type I and type II.

1. **Type I also known as IDDM (insulin dependent diabetes mellitus).** This is also known as juvenile onset diabetes mellitus. In this disease, insulin production is absent because of autoimmune β cellular destruction, possibly triggered by an environmental exposure in generally susceptible individuals. This disease usually develops in childhood or early adulthood.
2. **Type II DM is also known as NIDDM (non insulin dependent diabetes mellitus).** This is also known as an adult onset diabetes mellitus. In this disease, insulin secretion is inadequate. Often insulin levels are very high, especially early in the disease, but peripheral insulin resistance and increased hepatic production of glucose make insulin levels inadequate to normalize plasma glucose levels.

Symptoms and signs of DM

Symptoms and signs of DM are those of hyperglycaemia leading to urine frequency, polyuria and polydipsia that may produce orthostatic hypotension and dehydration. Severe dehydration causes weakness, fatigue and mental status changes. Symptoms may come and go as plasma glucose levels fluctuate. Polyphagia may accompany symptoms of hyperglycaemia but is not typically a primary concern. Hyperglycaemia can also cause weight loss, nausea and vomiting, blurred vision and it may predispose to bacterial and fungal infection.

1. **Patients with type 1 DM** present with symptoms of hyperglycaemia and sometimes diabetic ketoacidosis. Some patients experience a long, transient phase of near normal glucose levels after acute onset of the disease (honeymoon phase) due to partial recovery of insulin secretion.
 2. **Patients with type 2 DM** may present with symptomatic hyperglycaemia but are often asymptomatic, and their condition is detected only on routine testing. In some patients, initial symptoms are those of diabetic complications suggesting that the disease has been present for some time. In some patients, hyperosmotic coma occurs initially, especially during a period of stress or when glucose metabolism is impaired by drugs.
- Years of poorly controlled hyperglycaemia lead to multiple, primary vascular complications that affect small (microvascular), large (macrovascular) vessels or both. The mechanisms by which vascular disease

develops include glycosylation of serum and tissue proteins with formation of advanced glycation end products.

- Microvascular complications include **retinopathy, nephropathy and neuropathy.**

1. **Diabetic retinopathy:** This is the most common cause of adult blindness, it is characterized by retinal capillary microaneurysms initially and later by macular oedema and neovascularization. There are no early symptoms or signs, but focal blurring, vitreous or retinal detachment and partial or total vision loss eventually develop; rate of progression is highly variable. Diagnosis is done by retinal examination.
2. **Diabetic nephropathy:** One of the leading cause of chronic renal failure. It is characterized by thickening of the glomerular basement membrane, mesangial expansion and glomerular sclerosis. These changes cause glomerular hypertension and progressive decline in GFR. Systemic hypertension may accelerate progression. The disease is usually asymptomatic until nephrotic syndrome or renal failure develops. Diagnosis is by detection of urinary albumin.
3. **Diabetic neuropathy:** This is the result of nerve ischaemia due to microvascular disease, direct effects of hyperglycaemia on neurons, and intracellular metabolic changes that impair nerve function. Symmetric polyneuropathy, autonomic neuropathy, radiculopathy, cranial neuropathy and mononeuropathy are some of the variations observed in the patients of DM.
4. **Macrovascular disease:** Large vessel atherosclerosis is the result of hyperinsulinaemia, dyslipidaemia and hyperglycaemia. Manifestations are angina pectoris and MI, transient ischaemic attacks (TIA) and strokes; peripheral arterial disease. Diagnosis is done by history and examination.
5. **Cardiomyopathy:** diabetic cardiomyopathy is thought to result from many factors including epicardial atherosclerosis, hypertension and left ventricular hypertrophy, microvascular disease, endothelial and autonomic dysfunction, obesity and metabolic disturbances. Patients develop heart failure due to impairment in left ventricular systolic and diastolic function and more likely to develop heart failure after MI.
6. **Infection:** Patients with poorly controlled DM are prone to bacterial and fungal infections because of adverse effects of hyperglycaemia on granulocyte and T cell function. Most common are microcutaneous fungal infections (oral or vaginal candidiasis), bacterial foot infections (osteomyelitis), which are typically exacerbated by lower extremity vascular insufficiency and diabetic neuropathy.

7. **Other complications:** Diabetic foot complications (skin changes, ulceration, infection, gangrene) are common and are attributable to vascular disease, neuropathy and relative immunosuppression.

The following are the risk factors for the development of DM: Age is 35 years or older; patient having overweight or obesity; patient having a sedentary lifestyle; patient having the family history of DM; patient who is considered as having prediabetes; patient had diabetes during pregnancy or had a baby who weighed more than 4,000 gms at birth; patient who is having high blood pressure; patient who has a lipid disorder such as high cholesterol; the patient who had the history of cardiovascular disease; the patient who had steatotic liver disease (previously termed as fatty liver disease); the patient who had the history of polycystic ovary disease; the patient who has racial or ethnic ancestry that is associated with high risk and the patient who has the HIV infection. People with these risk factors should be screened for diabetes mellitus at least once every three years.

• **Diagnosis** is by fasting plasma glucose levels, assessment of HbA1C level. This is usually detected by doing the screening tests on people who are at risk of developing DM, but have no symptoms. These tests are performed in the patients who are having increased thirst, increased frequency of micturition, increased hunger. Some of the occasions, the patients may have frequent infections, foot ulcers and yeast infections.

• Ideally, the test is preferred by making the people to have fasting overnight. Diabetes can be diagnosed if the blood glucose levels are 126 mg/dl (7.0 mmol/L) or higher. However, it is possible to use blood samples taken after people have eaten. Some elevation of blood glucose levels after eating is normal, but even after a meal the levels should not be very high. Diabetes can be diagnosed if a random plasma glucose level is higher than 200 mg/dl.

• **HbA1C (Glycosylated haemoglobin):** This investigation reflects the long term trends in blood glucose levels rather than rapid changes. Haemoglobin is the red, oxygen carrying substance in red blood cells. When blood is exposed to high blood glucose levels over a long period of time, glucose attaches to the haemoglobin and forms glycosylated haemoglobin. The Haemoglobin A1C level blood test is reported as the percentage of haemoglobin that is A1C.

• Measurements of haemoglobin A1C can be used to diagnose diabetes when testing is done by a certified laboratory (not by instruments used at home or in a doctor's office). People with a HbA1C level of 6.5% or more have diabetes. If the level is between 5.7 and 6.4, they have prediabetes and are at risk of developing diabetes.

Pancreatitis

Pancreatitis can be classified as acute or chronic. Acute pancreatitis is the inflammation that resolves both clinically and histologically.

Chronic pancreatitis is characterized by histological changes that are irreversible and progressive and that result in considerable loss of exocrine and endocrine pancreatic function. Patients with chronic pancreatitis may have a flare up of acute disease.

- Pancreatic acinar cells secrete bicarbonate and digestive enzymes into the ducts that connect pancreas to the duodenum at ampulla of Vater (exocrine function). Pancreatic β cells secrete insulin directly into the blood stream (endocrine function). Both of these functions will be afflicted in the patients of pancreatitis.
- Biliary tract disease and alcoholism account for 80% of acute pancreatitis cases. The remaining 20% of the cases can occur because of mechanical causes such as gall stones, trauma, pancreatic or periampullary cancer; metabolic causes such as hypertriglyceridemia, hypercalcemia, oestrogen use; exposure to toxins such as alcohol, methanol; infectious diseases such as mumps; drugs intake such as ACE inhibitors, azathioprine, furosemide etc.
- An acute attack causes steady, boring upper abdominal pain, typically severe to require large doses of parenteral opioids. The pain observed in the epigastric region radiating towards the back through the left side.
- Pain usually develops suddenly in gallstone pancreatitis; and in alcoholic pancreatitis, the pain develops over a few days. The pain usually persists for several days. Sitting up and leaning forward may reduce pain, but coughing, vigorous movement, and deep breathing may accentuate it. Nausea and vomiting (projectile vomiting) are common.

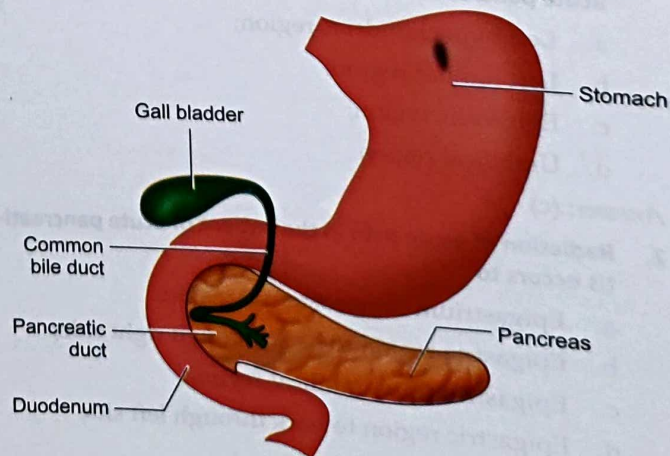


Figure 164: Pancreatitis

- The patient appears acutely ill and sweaty. Pulse rate is usually 100 to 140 beats/ minute. Respiration is shallow and rapid. BP may be transiently high or low, with significant postural hypotension. Temperature may be normal or even subnormal. Sensorium may be blunted to the point of semicoma. Scleral icterus is occasionally present. The lungs may have limited diaphragmatic excursion and evidence of atelectasis.
- Chronic pancreatitis is persistent inflammation of pancreas that results in permanent structural damage with fibrosis and ductal strictures, followed by a decline in exocrine and endocrine function. It can occur as the result of chronic alcohol abuse but may be idiopathic. Initial symptoms are recurrent attacks of pain. Late in the disease, some patients develop malabsorption and glucose intolerance.

- Type 3c diabetes develops when the pancreas of the individual experiences damage, such as chronic pancreatitis or cystic fibrosis. People with type 3c diabetes often also have a lack of pancreatic enzymes that are important for digestion.
- Diagnosis is done by analysis of serum markers (amylase and lipase). Once pancreatitis is diagnosed, CT is usually done. In chronic pancreatitis, the diagnosis is done by imaging studies such as ERCP, endoscopic ultrasound. Normal serum amylase should be between 15 – 110 IU/L should be present. Normal range for serum lipase for adults younger than 60 years is 10 to 140 U/L and in adults aged 60 years and above is 24 to 151 U/L. Higher than normal levels of lipase 3 to 10 times the normal levels of lipase, may imply the diagnosis of acute pancreatitis.

Key summary of the chapter

- Diabetes mellitus (DM) is impaired insulin secretion and variable degrees of peripheral insulin resistance leading to hyperglycaemia. Early symptoms are related to hyperglycaemia and include polydipsia (excessive thirst), polyphagia (excess eating) and polyuria (passage of excess amount of urine). Later complications include vascular diseases, peripheral neuropathy and predisposition to infection. Diagnosis is by measuring plasma glucose.
- Pancreatitis can be classified as acute or chronic. Acute pancreatitis is the inflammation that resolves both clinically and histologically. Chronic pancreatitis is characterized by histological changes that are irreversible and progressive and that result in considerable loss of exocrine and endocrine pancreatic function. Patients with chronic pancreatitis may have a flare up of acute disease.

Self-Assessment: Questions for Reviews

Short essay questions

- Explain the types of diabetes mellitus
- Explain type 2 diabetes mellitus
- Explain the types of pancreatitis
- Explain acute pancreatitis
- Describe pancreatitis induced diabetes mellitus

MCQs

1. The site of origin of abdominal pain in the patient of acute pancreatitis is

- a. Left hypochondriac region;
- b. Left lumbar region;
- c. Epigastric region;
- d. Umbilical region

Answer: (c)

2. Radiation of acute pain in the patient of acute pancreatitis occurs to

- a. Epigastrium to retrosternal region;
- b. Epigastric region to back through right side;
- c. Epigastrium to umbilical region;
- d. Epigastric region to back through left side

Answer: (d)

3. To diagnose the diabetes mellitus HbA1C level should be above

- a. 4;
- b. 5.5;
- c. 6;
- d. 6.5

Answer: (d)

4. Acquired onset of diabetes mellitus is also known as

- a. Juvenile diabetes mellitus;
- b. Type 1 diabetes mellitus;
- c. Type 2 diabetes mellitus;
- d. Type 3c diabetes mellitus

Answer: (c)

5. Type 2 diabetes mellitus is also known as

- a. Adult onset diabetes mellitus;
- b. Juvenile diabetes mellitus;
- c. Hereditary diabetes mellitus;
- d. None of the above

Answer: (a)

6. Which of the following test is indicative of average level of glucose in the past 3 months?

- a. Fasting plasma glucose assessment;
- b. Random plasma glucose assessment;

- c. HbA1C;
- d. All of the above

Answer: (c)

7. Diabetes mellitus is caused by the deficiency of production of

- a. Insulin;
- b. Glucagon;
- c. Lipase;
- d. Thyroxine

Answer: (a)

8. Which of the following feature should not be essentially be present in the patient of diabetes mellitus?

- a. Polyuria;
- b. Polyphagia;
- c. Polydipsia;
- d. Haematuria

Answer: (d)

9. The primary organ afflicted in the patient of diabetes mellitus is

- a. Pancreas;
- b. Liver;
- c. Thyroid gland;
- d. Adrenal gland

Answer: (a)

10. The diagnosis of acute pancreatitis is made when the serum amylase level raises above

- a. 30 mg/dl;
- b. 110 IU/L;
- c. 80 U/L;
- d. 1,000 IU/L

Answer: (b)

Chapter resources

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